Python · Quick concepts

Interview revision sheet. Read it the evening before, then test yourself in Interview practice.

Basics

- Types: int, float, str, bool, None. Collections: list, tuple, dict, set.
- Mutable: list, dict, set. Immutable: tuple, str, int.
- // is floor division, % is remainder, ** is power.
- Falsy values: 0, '', None, [], {}, set().
- f-strings: f"Hello {name}". Slicing: s[1:3], s[::-1] reverses.

Control flow and functions

- for x in items; range(3) gives 0, 1, 2; break exits, continue skips.
- enumerate gives (index, value); zip pairs items by position.
- *args collects extra positional args (tuple); **kwargs collects named args (dict).
- Do not use mutable defaults like def f(x=[]); use None and create inside.
- lambda x: x * 2 is a small anonymous function.

Comprehensions and generators

- List: [x * 2 for x in nums if x > 0]. Dict: {k: v for k, v in pairs}.
- Generator expression (x for x in data) is lazy and saves memory.
- yield turns a function into a generator.

Errors and resources

- try / except / else / finally; finally always runs.
- raise ValueError('message') to signal errors.
- with open('f.txt') as f: closes the file automatically.

OOP

- class, __init__ sets attributes, self is the instance.
- @staticmethod gets no self; @classmethod gets cls; @property makes a method look like an attribute.
- Dunder methods: __str__ (readable), __repr__ (developer), __len__, __eq__.
- @dataclass generates __init__, __repr__ and __eq__.

Advanced

- Decorator: a function that wraps another function, such as for logging or timing.
- Closure: an inner function remembers outer variables; nonlocal changes them.
- GIL: one thread runs Python bytecode at a time, so use multiprocessing for CPU work and threads or asyncio for I/O.
- is checks identity; == checks value. Use "is None".
- copy is shallow; deepcopy copies nested objects.

Useful modules

- collections: Counter, defaultdict, deque. functools: lru_cache.
- json.loads / json.dumps. requests.get(url, timeout=10).
- subprocess.run([...], capture_output=True, text=True) instead of os.system.
- logging instead of print for production scripts.

Project hygiene

- python -m venv .venv for isolated dependencies; pip freeze > requirements.txt.
- if __name__ == "__main__": runs code only when the file is executed directly.
- pytest for tests; type hints help tools like mypy.